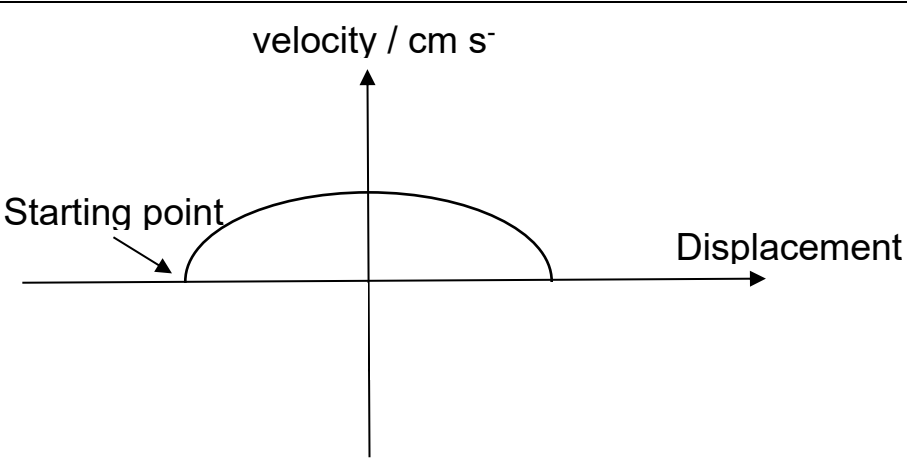


2a	An oscillatory motion where acceleration is directly proportional to displacement (from the equilibrium position), directed towards the equilibrium position.	1 1
2bi	$0.5T \Rightarrow 13 \text{ dots}$ $0.5T = 12 \times (1/50) = 0.24 \text{ s}$ $T = 0.48 \text{ s}$	1
2bii	$x_0 = 12.5 \text{ cm (first to last dot)} / 2 = 6.25 \text{ cm}$	1
2ci	$v_0 = \omega x_0 = \frac{2\pi}{T} x_0$ $v_0 = \frac{2\pi}{0.48} (6.25) = 81.8 \text{ cm s}^{-1}$	1
2cii	$F_0 = m\omega^2 x_0 = m \left(\frac{2\pi}{T} \right)^2 x_0$	1
	$F_0 = (0.700) \left(\frac{2\pi}{0.48} \right)^2 (0.0625) = 7.50 \text{ N}$	1
2d		2

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C2

2a	When two (or more) waves overlap, the resultant displacement (at any point and instant)	2
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